Inorganic Chemistry

Inorganic chemistry deals with synthesis and behavior of inorganic and organometallic compounds.

This field covers chemical compounds that are not carbon-based, which are the subjects of organic chemistry. The distinction between the two disciplines is far from absolute, as there is much overlap in the subdiscipline of organometallic chemistry. It has applications in every aspect of the chemical industry, including catalysis, materials science, pigments, surfactants, coatings, medications, fuels, and agriculture.

Occurrence

Many inorganic compounds are found in nature as minerals.[2] Soil may contain iron sulfide as pyrite or calcium sulfate as gypsum. [3] [4] Inorganic compounds are also found multitasking as biomolecules:
as electrolytes (sodium chloride), in energy storage (ATP) or in construction (the polyphosphate backbone in DNA)

Bonding

Inorganic compounds exhibit a range of bonding properties. Some are ionic compounds, consisting of very simple cations and anions joined by ionic bonding. Examples of salts (which are ionic compounds) are magnesium chloride $MgCl_2$, which consists of magnesium cations Mg^{2+} and chloride anions Cl^-; or sodium hydroxide $NaOH$, which consists of sodium cations Na^+ and hydroxide anions OH^-. Some inorganic compounds are highly covalent, such as sulfur dioxide and iron pentacarbonyl. Many inorganic compounds feature polar covalent bonding, which is a form of bonding intermediate between covalent and ionic bonding. This description applies to many oxides, carbonates, and halides.

Many inorganic compounds are characterized by high melting points. Some salts (e.g., NaCl) are very soluble in water.

When one reactant contains hydrogen atoms, a reaction can take place by exchanging protons in acid-base chemistry. In a more general definition, any chemical species capable of binding to electron pairs is called a Lewis acid; conversely any molecule that tends to donate an electron pair is referred to as a Lewis base.[5] As a refinement of acid-base interactions, the HSAB theory takes into account polarizability and size of ions.[5]

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Subdivisions of inorganic chemistry

Subdivisions of inorganic chemistry are numerous, but include:

- organometallic chemistry, compounds with metal-carbon bonds. This area touches on organic synthesis, which employs many organometallic catalysts and reagents.

- cluster chemistry, compounds with several metals bound together with metal–metal bonds or bridging ligands.

- bioinorganic chemistry, biomolecules that contain metals. This area touches on medicinal chemistry.

- materials chemistry and solid state chemistry, extended (i.e. polymeric) solids exhibiting properties not seen for simple molecules. Many practical themes are associated with these areas, including ceramics.

Industrial inorganic chemistry

Inorganic chemistry is a highly practical area of science. Traditionally, the scale of a nation's economy could be evaluated by their productivity of sulfuric acid.

An important man-made inorganic compound is ammonium nitrate, used for fertilization. The ammonia is produced through the Haber process. [6] [7] [8] Nitric acid is prepared from the ammonia by oxidation.

Another large-scale inorganic material is portland cement. Inorganic compounds are used as catalysts such as vanadium(V) oxide for the oxidation of sulfur dioxide and titanium(III) chloride for the polymerization of alkenes. Many inorganic compounds are used as reagents in organic chemistry such as lithium aluminium hydride.[9]